

Once we compute $\hat{U}_8^k$ and $V(\hat{U}_8^k)$ for $1 \leq k \leq K$, the estimator of U_8 is

$$\hat{U}_8 = \frac{\sum \hat{U}_8^k}{K},$$

and it's variance is

$$V(\hat{U}_8) = \frac{1}{K} \sum_{k=1}^K V(\hat{U}_8^k) + (1 + \frac{1}{K}) \sum_{k=1}^K \frac{(\hat{U}_8^k - \hat{U}_8)^2}{K-1}.$$

So the test statistic is

$$\frac{\hat{U}_8}{\sqrt{V(\hat{U}_8)}}.$$